

GEORGE PAULY

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PROFESSIONAL SUMMARY

- Software Engineer with experience in Python and Web Frameworks
- Built projects like HTTP 1.1 Server, AI Knowledge Base and LISP interpreter
- 3 years experience in data analysis and delivery leadership

TECHNICAL SKILLS

- Languages: Python, SQL, JavaScript
- Backend: FastAPI, Django, Flask, RESTful APIs, WebSockets.
- Databases & Tools: PostgreSQL, pgvector, Redis, MongoDB
- Frontend & Tools: React, CSS, HTML, uv, pip, Git, OpenRefine.

WORK EXPERIENCE

Delivery Lead – Data & Insights | Clootrack Software Labs | Dec 2023 – Jan 2025

- Led data delivery initiatives, ensuring accuracy and high-integrity dashboards and reports.

Data Analyst | Clootrack Software Labs | June 2022 – Nov 2023

- Extracted and cleaned large datasets using WebscraperIO, Python scripts and OpenRefine, maintaining data accuracy.

Data Intern | Clootrack Software Labs | Feb 2022 – May 2022

TECHNICAL PROJECTS

Custom HTTP 1.1 Web Server | Python, Sockets

- Built a web server using Python's socket library to implement the fundamentals of the HTTP/1.1 protocol
- Developed persistent connections and parsing for headers and body content according to RFC specs
- Implemented Trie-based routing logic to achieve optimal efficiency

AI Knowledge Base | FastAPI, PostgreSQL (pgvector), Embedding models and LLMs

- Built a RAG (Retrieval-Augmented Generation) system to perform semantic search over podcast show notes.
- Utilized pgvector for storing and querying high-dimensional embeddings generated by the Qwen 3 model.
- Hybrid search (Vector + Text) and integrated OpenRouter API for context-aware query responses.

Scheme (Lisp) Interpreter | Python

- Implemented core features including lexical scoping and lambda expressions

Real-Time Chat Application | React, FastAPI, Redis, WebSockets

- Built a bi-directional messaging platform using WebSockets for instant data transmission.
- Managed secure user sessions and authentication using Redis for session key storage and Cookie-based login

EDUCATION

- Software Engineering Bootcamp | Bit by Bit
- Master of Science (M.Sc.) in Mathematics | Mysore University
- Bachelor of Science (B.Sc.) in Mathematics, Physics, and Electronics | Christ University